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July 25, 2025

Assignment Research: Report 2 Wireshark

What did you do?

I captured my network traffic with Wireshark (version 4.4.8), a free, open-source packet analyzer. After reviewing the provided materials and doing additional research, I ran a packet capture for about three minutes. I first inspected the raw output and took a screenshot of it. Then I used the Statistics tab to see which protocols and hosts dominated the traffic.

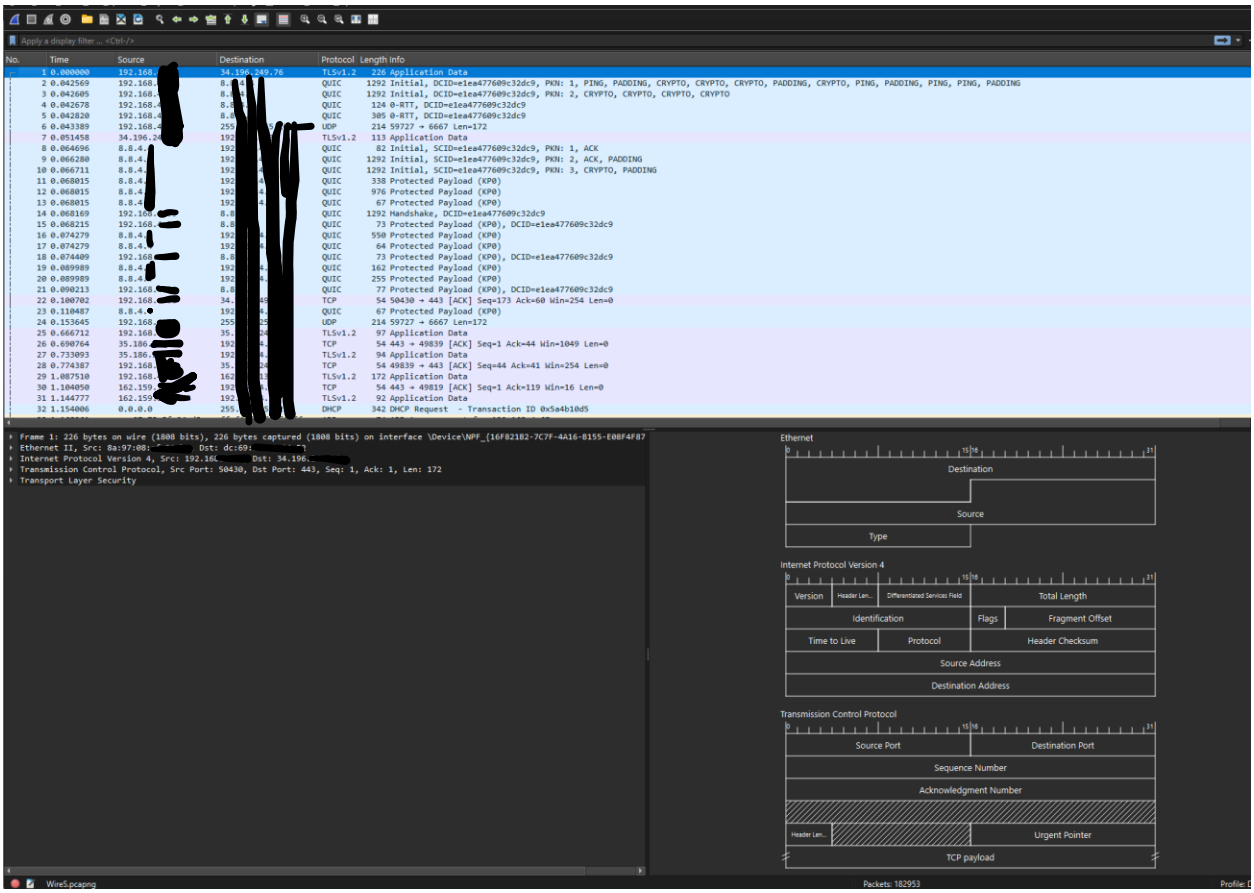
I began my analysis with Protocol Hierarchy to see which protocols were present. My next step was examining the Conversations views for Ethernet, IPv4, TCP, and UDP to understand who talked to whom and how much data moved. Applying display filters while toggling among these panes gave me a broader picture of what traverses my network.

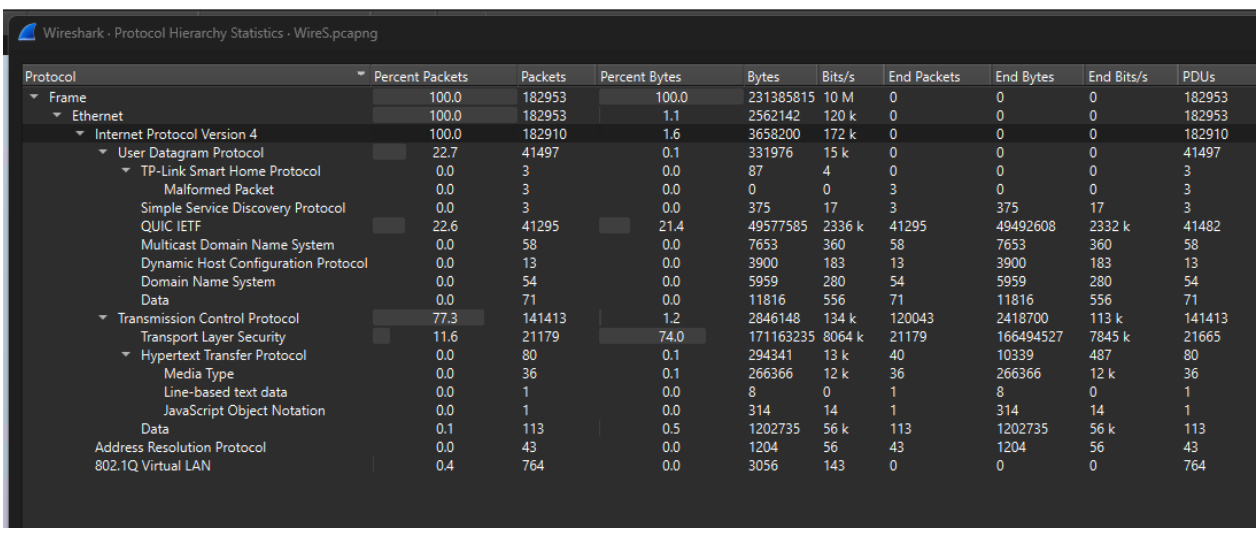
Finally, I checked Capture File Properties for metadata about the file. I will include a 19,630-page PDF of the collapsed output and images in the results section.

***Note: Expanded packed details could not be included in the PDF due to its size.

What are the results?

Some screenshots show only partial data to highlight the key points. The initial screenshot includes the packet diagram view in the bottom right.





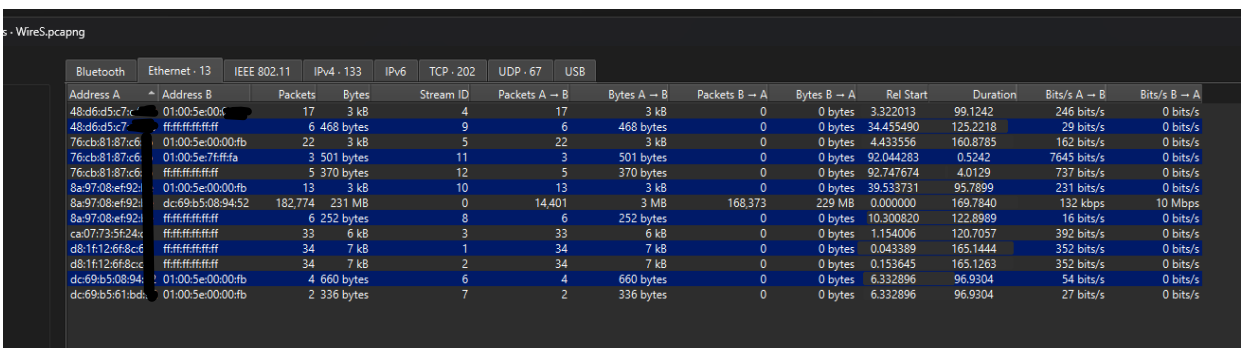
Wireshark · Protocol Hierarchy Statistics · WireS.pcapng

Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Packets	End Bytes	End Bits/s	PDUs
Frame	100.0	182953	100.0	231385815	10 M	0	0	0	182953
Ethernet	100.0	182953	1.1	2562142	120 k	0	0	0	182953
Internet Protocol Version 4	100.0	182910	1.6	3658200	172 k	0	0	0	182910
User Datagram Protocol	22.7	41497	0.1	331976	15 k	0	0	0	41497
TP-Link Smart Home Protocol	0.0	3	0.0	87	4	0	0	0	3
Malformed Packet	0.0	3	0.0	0	0	3	0	0	3
Simple Service Discovery Protocol	0.0	3	0.0	375	17	3	375	17	3
QUIC IETF	22.6	41295	21.4	49577585	2336 k	41295	49492608	2332 k	41482
Multicast Domain Name System	0.0	58	0.0	7653	360	58	7653	360	58
Dynamic Host Configuration Protocol	0.0	13	0.0	3900	183	13	3900	183	13
Domain Name System	0.0	54	0.0	5959	280	54	5959	280	54
Data	0.0	71	0.0	11816	556	71	11816	556	71
Transmission Control Protocol	77.3	141413	1.2	2846148	134 k	120043	2418700	113 k	141413
Transport Layer Security	11.6	21179	74.0	171163235	8064 k	21179	166494527	7845 k	21665
Hypertext Transfer Protocol	0.0	80	0.1	294341	13 k	40	10339	487	80
Media Type	0.0	36	0.1	266366	12 k	36	266366	12 k	36
Line-based text data	0.0	1	0.0	8	0	1	8	0	1
JavaScript Object Notation	0.0	1	0.0	314	14	1	314	14	1
Data	0.1	113	0.5	1202735	56 k	113	1202735	56 k	113
Address Resolution Protocol	0.0	43	0.0	1204	56	43	1204	56	43
802.1Q Virtual LAN	0.4	764	0.0	3056	143	0	0	0	764

Protocol Hierarchy:

- There were a total of 182,953 packets.
- Traffic is almost entirely Ipv4 over Ethernet.

- There is a lot of encrypted traffic, TLS with 21,179 packets.
- QUIC over UDP: 41,295 of the 41,497 UDP packets (also encrypted).
- Only 87 plain HTTP packets; most traffic is HTTPS (TLS) or HTTP/3 over QUIC.
- DNS is only 54 packets.
- Local-service chatter: DHCP, SSDP, mDNS, ARP are a total of 117 packets.
- There were 3 malformed packets from the TP-Link.
- There were 764 802.1Q tagged VLAN packets.



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Bluetooth | Ethernet · 13 | IEEE 802.11 | IPv4 · 133 | IPv6 | TCP · 202 | UDP · 67 | USB

Address A	Address B	Packets	Bytes	Stream ID	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A	Rel Start	Duration	Bits/s A → B	Bits/s B → A
48:d6:d5:c7:c[redacted]	01:00:5e:00:[redacted]	17	3 kB	4	17	3 kB	0	0 bytes	3.322013	99.1242	246 bits/s	0 bits/s
48:d6:d5:c7[redacted]	ff:ff:ff:ff:ff:ff	6	468 bytes	9	6	468 bytes	0	0 bytes	34.455490	125.2218	29 bits/s	0 bits/s
76:cb:81:87:c6[redacted]	01:00:5e:00:00:fb	22	3 kB	5	22	3 kB	0	0 bytes	4.433556	160.8785	162 bits/s	0 bits/s
76:cb:81:87:c6[redacted]	01:00:5e:7f:ff:fa	3	501 bytes	11	3	501 bytes	0	0 bytes	92.044283	0.5242	7645 bits/s	0 bits/s
76:cb:81:87:c6[redacted]	ff:ff:ff:ff:ff:ff	5	370 bytes	12	5	370 bytes	0	0 bytes	92.747674	4.0129	737 bits/s	0 bits/s
8a:97:08:ef:92[redacted]	01:00:5e:00:00:fb	13	3 kB	10	13	3 kB	0	0 bytes	39.533731	95.7899	231 bits/s	0 bits/s
8a:97:08:ef:92[redacted]	dc:69:b5:08:94:52	182,774	231 MB	0	14,401	3 MB	168,373	229 MB	0.000000	169.7840	132 kbps	10 Mbps
8a:97:08:ef:92[redacted]	ff:ff:ff:ff:ff:ff	6	252 bytes	8	6	252 bytes	0	0 bytes	10.300820	122.8989	16 bits/s	0 bits/s
ca:07:73:5f:24[redacted]	ff:ff:ff:ff:ff:ff	33	6 kB	3	33	6 kB	0	0 bytes	1.154006	120.7057	392 bits/s	0 bits/s
d8:1f:12:6f:8c:6[redacted]	ff:ff:ff:ff:ff:ff	34	7 kB	1	34	7 kB	0	0 bytes	0.043389	165.1444	352 bits/s	0 bits/s
d8:1f:12:6f:8c:c[redacted]	ff:ff:ff:ff:ff:ff	34	7 kB	2	34	7 kB	0	0 bytes	0.153645	165.1263	352 bits/s	0 bits/s
dc:69:b5:08:94[redacted]	01:00:5e:00:00:fb	4	660 bytes	6	4	660 bytes	0	0 bytes	6.332896	96.9304	54 bits/s	0 bits/s
dc:69:b5:61:bd[redacted]	01:00:5e:00:00:fb	2	336 bytes	7	2	336 bytes	0	0 bytes	6.332896	96.9304	27 bits/s	0 bits/s

Ethernet Conversation:

- The largest packet flow was between Address A at **:**:**:**:**:** and Address B at **:**:**:**:**:**.
- Total bytes exchanged: ~231 MB; the second and third largest flows were ~7 kB each.
- Most data moved from B to A.
- Broadcast and multicast traffic appeared normal.

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Bluetooth | Ethernet · 13 | IEEE 802.11 | IPv4 · 133 | IPv6 | TCP · 202 | UDP · 67 | USB

Address A	Address B	Packets	Bytes	Stream ID	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A	Rel Start	Duration	Bits/s A → B	Bits/s B → A
0.0.0.0	255.255.[redacted]	13	4 kB	6	13	4 kB	0	0 bytes	1.154006	119.0199	298 bits/s	0 bits/s
8.8.8.8	192.168.[redacted]	73	17 kB	19	39	11 kB	34	6 kB	9.512407	144.5346	610 bits/s	313 bits/s
18.211.[redacted]	192.168.[redacted]	79	7 kB	8	34	3 kB	45	4 kB	1.982832	165.8865	164 bits/s	180 bits/s
34.204.[redacted]	192.168.[redacted]	4	247 bytes	32	2	139 bytes	2	108 bytes	20.735221	40.0242	27 bits/s	21 bits/s
44.215.[redacted]	192.168.[redacted]	4	247 bytes	31	2	139 bytes	2	108 bytes	19.939177	40.0369	27 bits/s	21 bits/s
52.108.2[redacted]	192.168.[redacted]	22	2 kB	20	11	957 bytes	11	594 bytes	10.273015	157.0508	48 bits/s	30 bits/s
52.108.2[redacted]	192.168.[redacted]	42	3 kB	10	21	2 kB	21	1 kB	3.661543	158.0397	92 bits/s	57 bits/s
172.64.1[redacted]	192.168.[redacted]	24	2 kB	29	16	1 kB	8	656 bytes	16.350424	140.0075	60 bits/s	37 bits/s
192.16[redacted]	224.0.0.2[redacted]	4	660 bytes	13	4	660 bytes	0	0 bytes	6.332896	96.9304	54 bits/s	0 bits/s
192.16[redacted]	224.0.0.2[redacted]	22	3 kB	11	22	3 kB	0	0 bytes	4.433556	160.8785	162 bits/s	0 bits/s
192.16[redacted]	239.255.[redacted]	3	501 bytes	114	3	501 bytes	0	0 bytes	92.044283	0.5242	7645 bits/s	0 bits/s
192.16[redacted]	224.0.0.2[redacted]	17	3 kB	9	17	3 kB	0	0 bytes	3.322013	99.1242	246 bits/s	0 bits/s
192.16[redacted]	255.255.[redacted]	6	468 bytes	66	6	468 bytes	0	0 bytes	34.455490	125.2218	29 bits/s	0 bits/s
192.16[redacted]	3.33.161.[redacted]	85	29 kB	88	35	6 kB	50	22 kB	40.992307	126.3714	407 bits/s	1396 bits/s
192.16[redacted]	3.87.150.[redacted]	23	11 kB	115	11	5 kB	12	5 kB	99.235873	60.2470	726 bits/s	713 bits/s
192.16[redacted]	3.161.136[redacted]	47	23 kB	51	22	7 kB	25	16 kB	28.277560	129.0304	447 bits/s	1001 bits/s
192.16[redacted]	3.161.150[redacted]	29	9 kB	60	14	3 kB	15	6 kB	30.503180	118.5586	197 bits/s	408 bits/s
192.16[redacted]	3.161.242[redacted]	27	9 kB	78	13	3 kB	14	6 kB	40.537522	116.7547	197 bits/s	406 bits/s
192.16[redacted]	3.161.242[redacted]	33	10 kB	79	17	3 kB	16	6 kB	40.537659	116.7551	238 bits/s	438 bits/s
192.16[redacted]	3.168.24.[redacted]	29	11 kB	73	14	4 kB	15	7 kB	40.283608	117.0086	265 bits/s	501 bits/s
192.16[redacted]	3.210.47.[redacted]	10	709 bytes	25	6	447 bytes	4	262 bytes	11.143982	52.9830	67 bits/s	39 bits/s
192.168[redacted]	3.218.0.1[redacted]	19	8 kB	119	8	2 kB	11	6 kB	129.417249	0.3523	41 kbps	129 kbps
192.168[redacted]	3.223.101[redacted]	23	10 kB	116	11	5 kB	12	6 kB	99.769433	60.2392	601 bits/s	751 bits/s
192.168[redacted]	3.233.158[redacted]	8	508 bytes	61	4	217 bytes	4	291 bytes	31.071107	15.0470	115 bits/s	154 bits/s
192.168[redacted]	8.6[redacted]	61	22 kB	1	27	9 kB	34	13 kB	0.042569	138.3954	520 bits/s	773 bits/s

IPv4 Conversation:

- 120 of 133 IPv4 conversations were between 192.168.***.*** (client) and the Internet.

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Bluetooth | Ethernet · 13 | IEEE 802.11 | IPv4 · 133 | IPv6 | TCP · 202 | UDP · 67 | USB

Address A	Port A	Address B	Port B	Packets	Bytes	Stream ID	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A	Rel Start	Duration	Bits/s A → B	Bits/s B → A	Flows
192.16[redacted]	50601	34.107[redacted]	80	32	2 kB	61	17	1 kB	15	1 kB	30.176771	116.0139	87 bits/s	81 bits/s	2
192.16[redacted]	50839	199.23[redacted]	80	362	311 kB	186	92	14 kB	270	297 kB	133.036923	1.1133	100 kbps	2134 kbps	72
192.16[redacted]	50846	199.23[redacted]	80	7	1 kB	190	4	583 bytes	3	916 bytes	134.500347	0.0838	55 kbps	87 kbps	2
192.16[redacted]	50848	199.23[redacted]	80	887	1 MB	192	94	7 kB	793	1 MB	134.597437	30.3607	1722 bits/s	290 kbps	2
192.16[redacted]	50849	199.23[redacted]	80	157	165 kB	193	35	3 kB	122	163 kB	134.599196	30.3597	698 bits/s	42 kbps	2
192.16[redacted]	49703	104.20[redacted]	443	2	121 bytes	177	1	55 bytes	1	66 bytes	101.589104	0.0420	10 kbps	12 kbps	1
192.16[redacted]	49783	162.15[redacted]	443	1,781	382 kB	6	925	129 kB	856	253 kB	5.969494	144.0575	7181 bits/s	14 kbps	128
192.16[redacted]	49819	162.15[redacted]	443	57	6 kB	2	28	2 kB	29	3 kB	1.087510	165.0946	118 bits/s	149 bits/s	14
192.16[redacted]	49839	35.186[redacted]	443	24	2 kB	1	12	906 bytes	12	888 bytes	0.666712	150.0890	48 bits/s	47 bits/s	12
192.16[redacted]	49852	173.16[redacted]	443	125	38 kB	8	56	32 kB	69	6 kB	7.465225	160.1640	1601 bits/s	275 bits/s	27
192.16[redacted]	49910	104.20[redacted]	443	12	2 kB	34	8	604 bytes	4	912 bytes	25.649653	135.0965	35 bits/s	54 bits/s	8
192.16[redacted]	50056	104.18[redacted]	443	9	551 bytes	3	4	218 bytes	5	333 bytes	1.321574	84.7836	20 bits/s	31 bits/s	2
192.16[redacted]	50068	104.18[redacted]	443	7	430 bytes	36	3	163 bytes	4	267 bytes	26.152453	39.8077	32 bits/s	53 bits/s	2
192.16[redacted]	50104	199.23[redacted]	443	8	484 bytes	70	4	220 bytes	4	264 bytes	31.888282	135.0936	13 bits/s	15 bits/s	1
192.16[redacted]	50221	52.109[redacted]	443	12	816 bytes	10	6	492 bytes	6	324 bytes	8.409062	150.1458	26 bits/s	17 bits/s	1
192.16[redacted]	50241	13.225[redacted]	443	9	551 bytes	7	4	218 bytes	5	333 bytes	6.390232	59.8553	29 bits/s	44 bits/s	2
192.16[redacted]	50252	13.107[redacted]	443	37	22 kB	14	18	8 kB	19	14 kB	10.440976	150.8734	397 bits/s	751 bits/s	12
192.16[redacted]	50253	13.107[redacted]	443	40	22 kB	9	18	7 kB	22	14 kB	8.347571	151.2835	392 bits/s	759 bits/s	12
192.16[redacted]	50257	52.109[redacted]	443	12	816 bytes	26	6	492 bytes	6	324 bytes	15.679137	150.1244	26 bits/s	17 bits/s	1
192.16[redacted]	50259	13.107[redacted]	443	42	22 kB	11	18	7 kB	24	15 kB	8.613345	150.7299	376 bits/s	770 bits/s	12
192.16[redacted]	50266	52.109[redacted]	443	12	816 bytes	28	6	492 bytes	6	324 bytes	19.012456	150.1170	26 bits/s	17 bits/s	1
192.16[redacted]	50277	52.109[redacted]	443	10	680 bytes	35	5	410 bytes	5	270 bytes	25.869078	120.1506	27 bits/s	17 bits/s	1

TCP Conversation:

- Nearly everything was TCP/443 (TLS/HTTPS).
- Client 192.168.***.*** communicated with many CDN/Cloud IPs.
- Most streams showed significantly more bytes from Address B to Address A
- A few small exceptions, such as port 80

WireS.pcapng

Bluetooth | Ethernet · 13 | IEEE 802.11 | IPv4 · 133 | IPv6 | TCP · 202 | UDP · 67 | USB

Address A	Port A	Address B	Port B	Packets	Bytes	Stream ID	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A	Rel Start	Duration	Bits/s A → B	Bits/s B → A	Flows
8.8.	443	192.16	49975	4	289 bytes	12	2	181 bytes	2	108 bytes	9.512407	0.0233	62 kbps	37 kbps	1
18.2	443	192.16	50136	79	7 kB	4	34	3 kB	45	4 kB	1.982832	165.8865	164 bits/s	180 bits/s	34
34.1	443	192.16	50284	8	583 bytes	176	5	361 bytes	3	222 bytes	100.926740	0.0509	56 kbps	34 kbps	2
34.1	443	192.16	50270	7	529 bytes	178	4	307 bytes	3	222 bytes	105.383105	0.0511	48 kbps	34 kbps	2
34.1	443	192.16	50278	8	571 bytes	187	5	355 bytes	3	216 bytes	133.733716	0.0505	56 kbps	34 kbps	2
34.2	443	192.16	50238	4	247 bytes	31	2	139 bytes	2	108 bytes	20.735221	40.0242	27 bits/s	21 bits/s	1
44.2	443	192.16	50249	4	247 bytes	30	2	139 bytes	2	108 bytes	19.939177	40.0369	27 bits/s	21 bits/s	1
52.1	443	192.16	50265	22	2 kB	13	11	957 bytes	11	594 bytes	10.273015	157.0508	48 bits/s	30 bits/s	1
52.1	443	192.16	50294	22	2 kB	5	11	957 bytes	11	594 bytes	3.661543	158.0397	48 bits/s	30 bits/s	1
52.1	443	192.16	50269	20	1 kB	23	10	870 bytes	10	540 bytes	11.990526	142.0437	48 bits/s	30 bits/s	1
172.	443	192.16	50072	24	2 kB	27	16	1 kB	8	656 bytes	16.350424	140.0075	60 bits/s	37 bits/s	16
192.	50705	3.33.1	443	35	8 kB	114	16	3 kB	19	5 kB	40.992307	55.2060	439 bits/s	653 bits/s	8
192.	50707	3.33.1	443	50	21 kB	116	19	3 kB	31	18 kB	41.157545	126.2062	215 bits/s	1112 bits/s	13
192.	50817	3.87.1	443	23	11 kB	174	11	5 kB	12	5 kB	99.235873	60.2470	726 bits/s	713 bits/s	8
192.	50580	3.161	443	47	23 kB	50	22	7 kB	25	16 kB	28.277560	129.0304	447 bits/s	1001 bits/s	14
192.	50606	3.161	443	29	9 kB	66	14	3 kB	15	6 kB	30.503180	118.5586	197 bits/s	408 bits/s	8
192.	50683	3.161	443	27	9 kB	97	13	3 kB	14	6 kB	40.537522	116.7547	197 bits/s	406 bits/s	8
192.	50684	3.161	443	33	10 kB	98	17	3 kB	16	6 kB	40.537659	116.7551	238 bits/s	438 bits/s	10
192.	50655	3.168	443	29	11 kB	89	14	4 kB	15	7 kB	40.283608	117.0086	265 bits/s	501 bits/s	8
192.	50478	3.210	443	10	709 bytes	19	6	447 bytes	4	262 bytes	11.143982	52.9830	67 bits/s	39 bits/s	3
192.	50831	3.218	443	19	8 kB	182	8	2 kB	11	6 kB	129.417249	0.3523	41 kbps	129 kbps	8
192.	50818	3.223	443	23	10 kB	175	11	5 kB	12	6 kB	99.769433	60.2392	601 bits/s	751 bits/s	8
192.	50534	3.233	443	8	508 bytes	68	4	217 bytes	4	291 bytes	31.071107	15.0470	115 bits/s	154 bits/s	2
192.	50855	13.69	443	30	13 kB	195	18	8 kB	12	5 kB	135.842903	0.8210	73 kbps	52 kbps	6
192.	50809	13.89	443	39	33 kB	173	24	25 kB	15	8 kB	73.660478	0.1859	1077 kbps	334 kbps	6
192.	50252	13.10	443	37	22 kB	14	18	8 kB	19	14 kB	10.440976	150.8734	397 bits/s	751 bits/s	12
192.	50253	13.10	443	40	22 kB	9	18	7 kB	22	14 kB	8.347571	151.2835	392 bits/s	759 bits/s	12
192.	50259	13.10	443	42	22 kB	11	18	7 kB	24	15 kB	8.613345	150.7299	376 bits/s	770 bits/s	12
192.	50808	13.22	443	34	11 kB	172	16	4 kB	18	7 kB	69.350287	68.2954	469 bits/s	802 bits/s	9
192.	50241	13.22	443	9	551 bytes	7	4	218 bytes	5	333 bytes	6.390232	59.8553	29 bits/s	44 bits/s	2
192.	50564	13.226	443	30	11 kB	44	15	4 kB	15	7 kB	27.220629	117.8624	263 bits/s	496 bits/s	8
192.	50549	18.155	443	24	6 kB	32	13	4 kB	11	2 kB	22.344365	117.7223	272 bits/s	104 bits/s	8
192.	50474	18.160	443	10	681 bytes	21	6	426 bytes	4	255 bytes	11.144019	52.9671	64 bits/s	38 bits/s	3
192.	50475	18.160	443	10	681 bytes	20	6	426 bytes	4	255 bytes	11.144002	52.9680	64 bits/s	38 bits/s	3
192.	50563	18.161	443	473	96 kB	43	221	42 kB	252	54 kB	27.194119	138.9729	2426 bits/s	3100 bits/s	60

UDP Conversation:

- UDP/443 (QUIC/HTTP3) traffic to multiple CDN IPs showed many ~9.6 kB packets, typical of MTU-sized QUIC frames.
- UDP/53 (DNS) to 8.8.***.***/8.8.***.*** was minimal.
- DHCP, mDNS, SSDP, TP-Link (9999/UDP) traffic was low-volume and normal for a home LAN.
- One notable flow: UDP/4851 to 216.167.***.*** (~5.6 MB each way). Likely game/VoIP traffic, but it's the only large nonstandard UDP port.

Capture File Properties (additional capture details).

What did you learn?

I learned what network packet analysis is and how Wireshark implements it. A packet analyzer captures live packets via the network interface card (NIC), decodes them layer by layer, and presents human-readable fields that can be filtered. Wireshark's built-in Statistics tools reveal protocol breakdowns, top talkers, flow graphs, and more. I appreciated Wireshark's color rules that highlight traffic worth investigating. Wireshark can diagnose slow services, identify suspicious traffic (malware, scans), and verify whether applications send the correct messages. I learned all about this by using Wireshark's interface.

From my network results, I confirmed that most of my traffic is encrypted Ethernet data. The client system (my computer) was streaming video, gaming, and browsing. I saw that the broadcast/multicast traffic looked normal. Traffic primarily went to well-known services like Google, YouTube, and Twitch. Gaming applications created a large UDP flow, and the Discord app generated a bidirectional UDP stream. Overall, the capture indicates that I have a modern, security-conscious system using HTTPS and HTTP/3.

I did not see clear indicators of an active cyber attack when doing my analysis. There were no scan patterns, no massive outbound exfiltration, and no obvious malicious protocols. I learned there are still attack surface areas that can be improved/hardened. Encrypted traffic can hide threats, and high-volume UDP without handshakes can be abused if unmonitored. I should ensure only approved apps run. My IoT devices expand the attack surface due to weak credentials and firmware issues. Heavy reliance on CDNs/cloud services means an attacker could compromise or spoof them. I also learned that capturing traffic can expose passwords and personal data if the traffic is unencrypted.

For security planning and risk assessment, Wireshark can validate policies, watch for risky behavior, and help prioritize controls. Wireshark can help build a network baseline of

normal traffic to spot anomalies, such as large uploads to unknown hosts or unencrypted clear-text data. Wireshark's detailed packet views can document an intrusion and provide evidence. This dataset will help me make smarter security and operational decisions in the future. By saving summaries and comparing over time, I can detect rare events faster (e.g., large uploads), find blind spots, and rely on timestamps for evidence. Each new capture increases the baseline's value.